Vance Vanvolkenburgh
655.662 — Introduction to Healthcare Systems Engineering
Module 7 | M7A1: Interface Requirements

Interface Requirements — Peritoneal Dialysis Remote Patient Monitoring (PD-RPM)

The five requirements below are allocated from the operational requirements and key performance parameters established in M3A1 and are written against the physical architecture and interface register of M5A1, so each one names an interface that already exists in the baseline. Each states a single **shall**, is verifiable by a named method, and identifies the interface explicitly: the two elements it joins, which side of the system boundary each sits on, the class of interface, and what crosses it. Three of the five govern interfaces whose far side is **outside** the PD-RPM boundary — the cycler, the carrier network, and the electronic health record — and those are stated as obligations on PD-RPM elements, never on parties the program does not control.

Req	Interface	Elements joined (M5A1 IDs)	Class	What crosses
IR-1	Cycler treatment-record acquisition	I-1 — PD Cycler (CY-1, external) → Home Gateway (HG-1)	Physical and data	Data
IR-2	Store-and-forward across the cellular link	I-5 — Home Gateway (HG-1) ↔ LTE-M Carrier Network (CN-1, external)	Data (wireless)	Signals, data
IR-3	Electronic health record exchange	I-16 — EHR Integration Service (EI-1) ↔ Electronic Health Record (EH-1, external)	Data (software)	Data
IR-4	Clinician triage presentation	I-13 — Clinician Web Application (CW-1) → Clinician Workstation (CS-1)	User	Data, signals
IR-5	Home electrical supply and ride-through	new (I-17) — Household mains (external) → Home Gateway (HG-1)	Physical	Energy

IR-1 — Cycler treatment-record acquisition · interface I-1 · Physical and data interface · PD Cycler (CY-1, external) → Home Gateway (HG-1)

The Home Gateway (HG-1) shall acquire the complete treatment record for each cycler session from the PD Cycler (CY-1) over a USB 2.0 full-speed (12 Mbit/s) Type-B data port within 5 minutes of session end, where a complete record comprises the fill volume, drain volume, dwell time, net ultrafiltration, and alarm codes for every cycle in that session, and shall reject and log any record failing its CRC-16 check without substituting a default value.

Allocated from: OR-2; KPP Data Completeness (95% of sessions ingested within 24 hours)
Verification: Test against a cycler simulator across all alarm codes, then demonstration against the production cycler on three consecutive overnight sessions

IR-2 — Store-and-forward across the cellular link · interface I-5 · Data (wireless) interface · Home Gateway (HG-1) ↔ LTE-M Carrier Network (CN-1, external)

The Home Gateway (HG-1) shall retain in non-volatile storage every treatment record, device measurement, and patient entry it has not successfully delivered to the LTE-M Carrier Network (CN-1), holding not fewer than 500 records or 72 hours of accumulated data, whichever is greater, shall declare the link unavailable after 3 consecutive failed transmission attempts within a 5-minute window, and shall deliver all retained items in chronological order within 30 minutes of link restoration.

Allocated from: OR-2; KPP Operational Availability ($A_o \geq 0.98$); risk R-1 (silent gateway) from M5A2
Verification: Test by injected link outages of 1, 24, and 72 hours in an RF-shielded enclosure, with record counts and ordering reconciled at the Edge Gateway (EG-1)

IR-3 — Electronic health record exchange · interface I-16 · Data (software) interface · EHR Integration Service (EI-1) ↔ Electronic Health Record (EH-1, external)

The EHR Integration Service (EI-1) shall exchange patient data with the Electronic Health Record (EH-1) using HL7 v2.5.1 ADT and ORU message types and FHIR R4 resources over TLS 1.3 with mutual certificate authentication, shall transmit each treatment summary within 15 minutes of clinician disposition, and shall retransmit any message for which an HL7 application acknowledgment is not received within 60 seconds, to a maximum of 3 attempts, before raising an integration fault to the monitoring center queue.

Allocated from: OR-4; the M2A2 boundary decision placing the EHR outside PD-RPM
Verification: Test against the EHR vendor conformance sandbox for every message type, including negative tests for withheld and malformed acknowledgments

IR-4 — Clinician triage presentation · interface I-13 · User interface · Clinician Web Application (CW-1) → Clinician Workstation (CS-1)

The Clinician Web Application (CW-1) shall render the complete triage queue on the Clinician Workstation (CS-1) within 3 seconds at the 95th percentile for a panel of up to 150 patients over a 10 Mbit/s connection, displaying for each queued patient the alert category, the triggering value with its units, the time elapsed since the triggering measurement, and the acknowledgment state, with no queue entry requiring horizontal scrolling at a display width of 1366 pixels.

Allocated from: OR-1 and OR-3; KPP Alert Latency (15 minutes to the monitoring nurse)
Verification: Demonstration against a 150-patient synthetic panel with timing captured by instrumented browser telemetry over 100 page loads

IR-5 — Home electrical supply and ride-through · interface new (I-17) · Physical interface · Household mains (external) → Home Gateway (HG-1)

The Home Gateway (HG-1) shall operate from a 100–127 VAC, 60 Hz single-phase household supply drawing no more than 15 W, shall sustain acquisition, buffering, and transmission at full function for not less than 24 hours on internal battery following loss of that supply, and shall transmit a loss-of-mains notification to the Device Management Server (DM-1) within 60 seconds of transferring to battery.

Allocated from: OR-2; KPP Operational Availability ($A_o \geq 0.98$)
Verification: Test by mains disconnection with the gateway under nominal session load, run to battery exhaustion, repeated at 15 °C and 30 °C ambient

Why the 24-hour figure in IR-5. The module 6 team-learning review traced a cluster of silent-device events to patients switching off a power strip overnight. A 6-hour gateway battery could not outlast a night-long session, and no single team held enough evidence to see it. IR-5 converts that finding into a bounded, testable obligation on the one element PD-RPM controls.

Three of these interfaces cross the system boundary and therefore cannot be closed by requirement alone. IR-1 depends on a signed Interface Agreement Document with the cycler manufacturer covering the port, the record schema, and notice of firmware changes; IR-2 depends on a carrier service agreement stating the LTE-M coverage and availability committed under the private APN; IR-3 depends on an Interface Control Document with the health system naming the HL7 message set, the FHIR profile version, and the acknowledgment semantics. Absent those three artifacts, the first risk on the module 7 interface risk list — lack of external interface commitments — is open against this design, and the requirements above are unverifiable in the field however precisely they are written.